

IAF/IAA SPACE LIFE SCIENCES SYMPOSIUM (A1)
Medical Care for Humans in Space (3)

Author: Mr. Adrian Tabari
University of British Columbia, Canada

Dr. Farshad Felfelian
University of British Columbia, Canada
Ms. Niyoosha Yoosefi
University of British Columbia, Canada
Dr. Mohammad Elahi
University of Toronto, Canada
Ms. Roxanna Tabari
University of British Columbia, Canada
Mr. Aaryamaan Verma
University of British Columbia, Canada
Dr. Matthew Turnock
University of British Columbia, Canada

THROMBOLYTIC PHARMACODYNAMICS IN MICROGRAVITY: IMPLICATIONS FOR
AUTONOMOUS MEDICAL CARE IN DEEP SPACE**Abstract**

Thrombosis represents a credible medical emergency during human spaceflight. Reduced internal jugular venous blood velocity, stasis, and retrograde venous flow in astronauts satisfy elements of Virchow's triad and increase the risk of deep vein thrombosis, pulmonary embolism, and cerebral sinus thrombosis. During future missions, communication delays and limited onboard medical resources will require crews to autonomously manage such events. Alteplase is the standard-of-care thrombolytic; however, the effects of reduced-gravity on drug-clot interface kinetics are unknown. Whether microgravity alters clot-dissolution dynamics remains to be evaluated, limiting the development of evidence-informed medical strategies for space operations.

This study investigates whether microgravity affects clot-dissolution by alteplase under controlled-flow conditions. We hypothesize that altered buoyancy-driven convection in microgravity modifies drug residence time at the clot interface, potentially influencing thrombolysis kinetics. To examine this, we developed a compact, flight-qualified, autonomous microfluidic platform housed within a single portable enclosure for parabolic flight operations.

Murine blood clots are formed in vitro within microfluidic channels by recalcifying citrated blood. The system incorporates parallel alteplase and saline perfusion arms, enabling contemporaneous experimental and control measurements under identical gravitational profiles. Precision peristaltic pumps deliver fluid at fixed infusion rates of 2.5 and 5.0 mL/min. A multiplexed electrically actuated pinch-valve manifold sequentially isolates individual clot channels, permitting controlled interrogation of each specimen within a single parabolic window. Differential pressure sensors positioned proximal and distal to each channel measure real-time pressure changes (ΔP) at >50 Hz, enabling time-resolved quantification of clot resistance decay during thrombolysis. High-resolution macro imaging provides complementary visual documentation of clot fragmentation. Pump control, valve sequencing, sensor acquisition, and time-stamped data logging are managed autonomously by an onboard single-board computer.

Preliminary ground studies confirmed stable flow regulation ($\pm 5\%$ of setpoint) and pressure sensitivity sufficient to resolve $>3\%$ of changes in clot-associated resistance. Alteplase-mediated thrombolysis demonstrated a statistically significant difference in ΔP at 20 seconds and in time-to-clot breakdown between 2.5 and 5.0 mL/min perfusion rates when adjusted for fluid-flow-mediated clot breakdown, consistent with flow-dependent drug exposure at the clot interface. These results validate the experimental architecture and support its sensitivity for detecting thrombolysis kinetics within the 20-second microgravity intervals provided by parabolic flight. Microgravity experiments are forthcoming through the Canadian Reduced Gravity Experiment Design Challenge (CAN-RGX) campaign.

This work establishes a quantitative framework for studying thrombolysis under reduced-gravity and supports future investigations of coagulation and fibrinolysis in spaceflight biomedical research.

Declaration on the use of Generative AI and AI-assisted Technologies in the writing process

To improve grammatical structure of the text.